



RNA-seq Gene Signatures Predict Survival in Breast Cancer.

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1. Introduction & Research Question

- Globally, breast cancer is the most diagnosed cancer in women (~2.3 million new cases/year)¹
- High molecular and phenotypic heterogeneity heavily impacts prognosis and treatment response.
- 4 molecular subtypes:** Lum. A, Lum. B, HER2-enriched, & Basal-like.
- Prognostic gene signatures can improve subtype classification and survival prediction.

Research Question:

- Can RNA-seq. multi-gene signatures from TCGA-BRCA data be used to predict overall survival in breast cancer patients?

2. Hypothesis & Prediction

Hypothesis:

- RNA-seq-based multi-gene signatures derived from subtype-specific differential expression in TCGA-BRCA patients are associated with overall survival and stratify patients into distinct prognostic risk groups.

Prediction:

- Patients with high-risk gene expression signatures will have shorter survival, while low-risk signatures predict longer survival.

3. Project Pipeline



4. Results

Figure 1. PCA plot of TCGA-BRCA samples by PAM50 subtype.

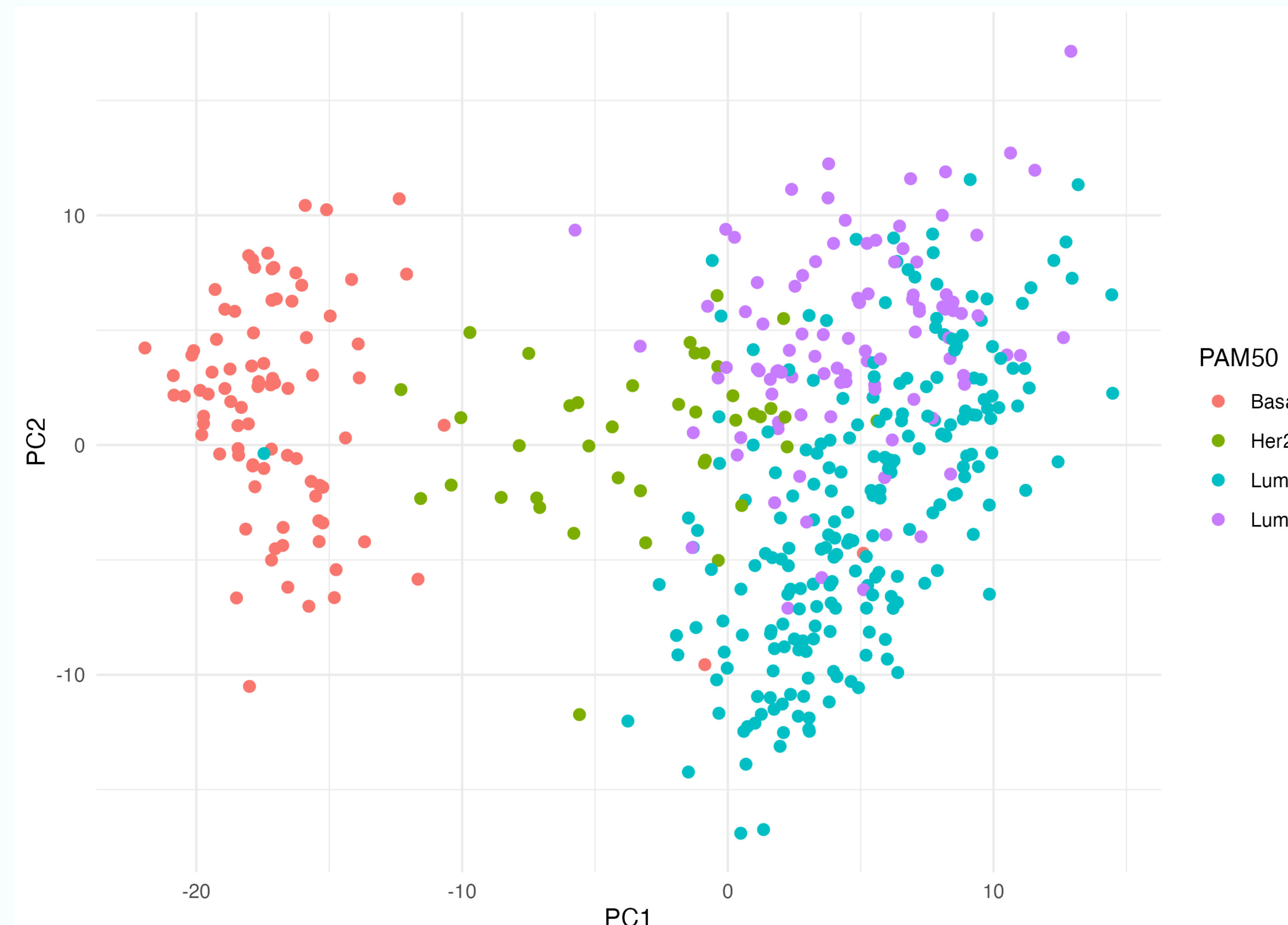
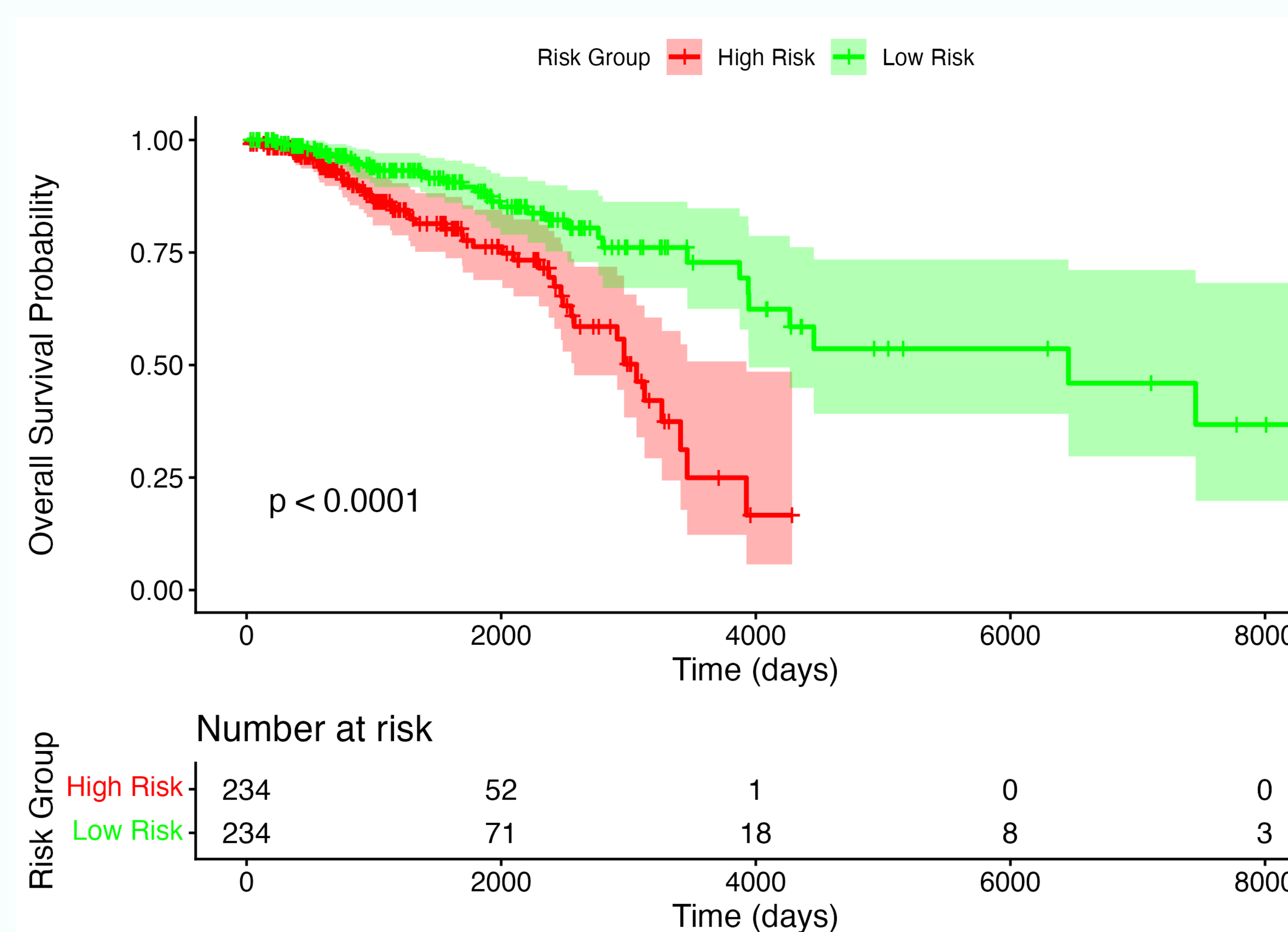


Figure 1. PCA of TCGA-BRCA PAM50 subtypes. Demonstrates distinct separation of Basal samples, significant overlap between LumA/LumB, and high heterogeneity in HER2.

Figure 2. Kaplan-Meier Survival Curves by Risk Group



- Pairwise Differential Expression:** 5,100 total DEGs across PAM50 subtypes, (notably, LumA vs. Basal yielded 4,348 DEGs, whereas LumB vs. HER2 yielded 1,565 DEGs).
- Each subtype has unique gene expression patterns and molecular signatures.
- The model identified a luminal gene signature (12 key upregulated genes) associated with overall survival - **BCL2, SERPINA3, GREB1, MAPT, FOXA1, CA12, TBC1D9, MLPH, SLC44A4, PARD6B, PPM1J, and LRFN2**.
- The model generated patient-specific risk scores to stratify patients into **High Risk** and **Low Risk** groups.
- MAPT + GREB1** (HR < 1) = strongest protective genes. Main functions = estrogen receptor signalling (GREB1 & MAPT), luminal differentiation (MAPT) → better prognosis²⁻⁵. Also features **BCL2** (estrogen receptor-positive/luminal phenotypic stability)^{6,7}.
- PARD6B + TBC1D9 + LRFN2 + FOXA1** (HR > 1) = moderate overall association with increased risk. Main functions = transcription regulation (FOXA1), epithelial organization/differentiation (PARD6B), signaling (TBC1D9) → tumor progression⁸⁻¹⁷.
- C-index** = 0.641 (se = 0.04) = **Moderate Predictive Ability**.
- Risk Score Hazard Ratio (HR) = 2.72 (95% CI: 1.88-3.93, p = 1.2x10⁻⁷)** (i.e., each unit increase in risk score = a 2.72-fold increase in the hazard of death).
- The signature risk score maintained independent prognostic value even after adjusting for key confounders such as age, tumor purity, and PAM50 subtype.
- Age** (r = -0.168) and **tumor purity** (r = -0.007) have weak/negligible direct correlations with overall survival.

Figure 2. Kaplan-Meier Survival Curves by Risk Group. The prognostic signature significantly stratified patients into high- and low-risk groups (**p < 0.0001**), with markedly **shorter median survival in the high-risk group (2,965 days)** compared with the **low-risk group (6,456 days)**.

5. Biological Interpretation, Conclusion & Future Work

- Gene expression differs across subtypes.
- Numerous prognostic gene signatures have already been established in breast cancer. While the gene signature identified in this analysis may be unique and contribute to prognosis, it builds upon an already existing concept.
- Single genes alone can't be used in prognosis.
- Generally, luminal cancers tend to have better prognosis than non-luminal cancers.
- This analysis supports the potential of RNA-seq data for precision oncology.
- Future studies** could focus on validating the survival model on an external dataset, and improving the model's accuracy and predictive performance by integrating more clinical variables (e.g., tumor purity), and increasing sample size.

6. References

- ¹(WHO, 2025); ²(Mohammed et al., 2013); ³(Makhlouf et al., 2025); ⁴(Wang et al., 2025); ⁵(Wang et al., 2019); ⁶(Hwang et al., 2012); ⁷(Hassan et al., 2025); ⁸(Nolan et al., 2008); ⁹(Catterall et al., 2020); ¹⁰(Cunliffe et al., 2012); ¹¹(Fonseca-Benitez et al., 2025); ¹²(Badve et al., 2007); ¹³(Hisamatsu et al., 2012); ¹⁴(Metovic et al., 2022); ¹⁵(Rangel et al., 2018); ¹⁶(De Lara et al., 2019); ¹⁷(Kothari et al., 2021); ¹⁸(Kothari et al., 2020)

Acknowledgements

I would like to thank Professor Nguyen for her invaluable constructive feedback throughout this project.